

INTRODUCTION

Universal Meals Program (UMP)

- Free meals for all K-12 public school students, regardless of income
- Implemented in CA during the 2022-23 academic year, making it the first state to pass a policy of this kind

Chronic Absenteeism

- Defined as missing 10% or more of the school year [1]
- Linked to lower academic performance, higher dropout rates, and increased delinquency risks [2, 3, 4, 7, 8]

Free Meals

- Reduce hunger-related absences [5]
- Improve attendance, class participation, and student health [6, 10]
- Criticized for increasing taxpayer costs, especially in wealthier districts with lower food insecurity

Can universal free meals deliver meaningful benefits across socioeconomic contexts?

DATA

Source

- California Department of Education (CDE)
 - Chronic absenteeism rates for K-12 public schools
 - Free and Reduced-Price Meals (FRPM) eligibility
- 7 academic years from 2016-17 to 2023-24
 - 2019-20 excluded due to missing data (COVID-19)
 - 2022-23 and 2023-24 treatment periods
- $N = 6,804$ across 972 districts
- District poverty levels defined by % on FRPM
 - Low ($\leq 25\%$), Mid-Low ($> 25\% - \leq 50\%$), Mid-High ($> 50\% - \leq 75\%$), High ($> 75\%$) [9]
 - Poverty level fixed on 2021-22 AY classification

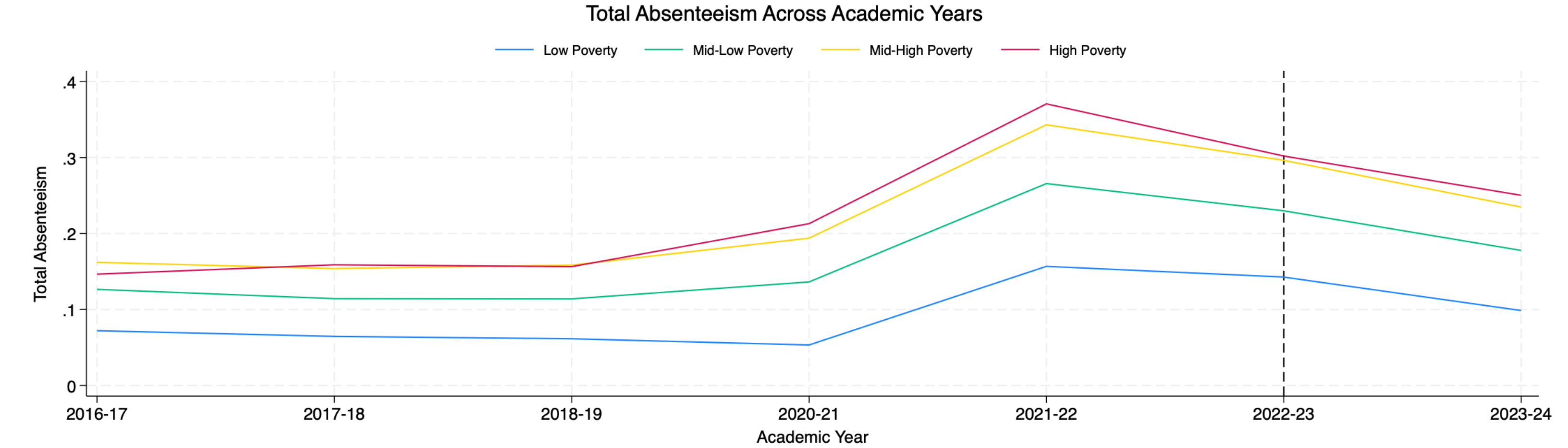
Summary Statistics

Chronic Absenteeism Rate by Demographic Across Academic Years								
	2016-17	2017-18	2018-19	2020-21	2021-22	2022-23	2023-24	Avg (%)
Female	.125	.127	.126	.145	.289	.245	.199	17.98%
Male	.127	.127	.129	.163	.293	.252	.204	18.54%
Asian	.059	.064	.063	.070	.162	.137	.110	9.54%
Black	.171	.175	.181	.220	.353	.304	.266	23.83%
Hispanic	.130	.132	.132	.171	.315	.269	.220	19.59%
White	.128	.129	.126	.137	.273	.238	.190	17.47%
Disabled	.174	.177	.178	.207	.369	.319	.265	24.21%
Foster	.256	.257	.253	.330	.435	.374	.352	31.88%
Homeless	.277	.284	.283	.347	.481	.424	.371	35.42%
Total	.127	.128	.127	.156	.293	.253	.203	18.43%

District Counts by Poverty Level Across Academic Years							
	2016-17	2017-18	2018-19	2020-21	2021-22	2022-23	2023-24
Low	198	147	151	163	153	136	120
Mid-Low	344	322	326	325	346	335	321
Mid-High	180	246	237	229	234	249	238
High	250	257	258	255	239	252	293
Total	972	972	972	972	972	972	972

RESULTS

Parallel Trends



Differential Impact

Chronic Absenteeism by Demographic									
	Gender		Race				Program Subgroup		
	Female	Male	Asian	Black	Hispanic	White	Disabled	Foster	Homeless
Post	0.0798*** (0.00484)	0.0818*** (0.00462)	0.0612*** (0.00630)	0.116*** (0.00835)	0.0922*** (0.00475)	0.0733*** (0.00531)	0.0968*** (0.00551)	0.0794*** (0.00950)	0.0945*** (0.00875)
Mid-Low Poverty	-0.00236 (0.00601)	0.00401 (0.00604)	0.0164* (0.00643)	0.0198* (0.00886)	0.00204 (0.00617)	0.00101 (0.00593)	0.00538 (0.00728)	0.00388 (0.00953)	0.000578 (0.00897)
Mid-High Poverty	-0.000117 (0.00824)	0.00398 (0.00806)	0.0279** (0.00949)	0.0273* (0.0133)	0.00460 (0.00840)	0.00295 (0.00888)	0.00718 (0.00961)	0.0134 (0.0153)	0.0411** (0.0143)
High Poverty	-0.00405 (0.0121)	-0.00275 (0.0105)	0.0273* (0.0113)	0.0253 (0.0156)	0.00147 (0.0111)	-0.00825 (0.0113)	0.00316 (0.0118)	0.00007 (0.0265)	0.0339 (0.0233)
Post · Mid-Low Poverty	0.00130 (0.00605)	0.00101 (0.00593)	-0.00668 (0.00773)	-0.0247* (0.0108)	0.000715 (0.00611)	-0.00710 (0.00656)	-0.00488 (0.00732)	0.0182 (0.0121)	0.0126 (0.0102)
Post · Mid-High Poverty	-0.00658 (0.00509)	-0.00612 (0.00514)	-0.00966 (0.00696)	-0.0260** (0.00897)	-0.00391 (0.00548)	-0.0155** (0.00547)	-0.00652 (0.00612)	0.0115 (0.0113)	0.00935 (0.00991)
Post · High Poverty	-0.0203*** (0.00565)	-0.0178** (0.00549)	-0.0232*** (0.00660)	-0.0202 (0.0109)	-0.00952 (0.00597)	-0.0231*** (0.00573)	-0.0119 (0.00630)	0.0637** (0.0232)	-0.00381 (0.0186)
Observations	6728	6754	4365	4129	6550	6578	6373	3676	4696
R-squared	0.511	0.518	0.332	0.407	0.542	0.444	0.528	0.281	0.336

Clustered by district. Standard errors in parentheses. District and year fixed effects included. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Summarized Model

	Total Absenteeism			
	(1)	(2)	(3)	(4)
Post	0.0658*** (0.00449)	0.0660*** (0.00450)	0.0770*** (0.00471)	0.0790*** (0.00475)
Mid-Low Poverty	-0.00386 (0.00666)	0.00832 (0.00791)	-0.00937 (0.00529)	-0.00194 (0.00597)
Mid-High Poverty	-0.0311*** (0.00699)	0.0136 (0.00997)	-0.0300*** (0.00618)	-0.000523 (0.00796)
High Poverty	-0.0823*** (0.00752)	-0.00114 (0.0123)	-0.0637*** (0.00764)	-0.00595 (0.0110)
Post · Mid-Low Poverty	-0.00159 (0.00663)	0.00144 (0.00665)	-0.000248 (0.00587)	0.00203 (0.00589)
Post · Mid-High Poverty	-0.0102 (0.00546)	-0.00617 (0.00539)	-0.00796 (0.00520)	-0.00494 (0.00517)
Post · High Poverty	-0.0214*** (0.00582)	-0.0176** (0.00576)	-0.0232*** (0.00554)	-0.0203*** (0.00550)
District Fixed Effect	No	Yes	No	Yes
Year Fixed Effects	No	No	Yes	Yes
Observations	6804	6804	6804	6804
R-squared	0.1615	0.107	0.3187	0.523

Clustered by district. Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

MODEL

Difference-in-Differences

$$\text{Absenteeism}_{it} = \beta_0 + \beta_1(\text{Post}_t) + \beta_2(\text{Post}_t \cdot \text{MidLow}_i) + \beta_3(\text{Post}_t \cdot \text{MidHigh}_i) + \beta_4(\text{Post}_t \cdot \text{High}_i) + \delta_i + \gamma_t + \epsilon_{it}$$

In the absence of a clear control, the *Low Poverty* group serves as a comparison; its near-zero, insignificant post-treatment effect suggests minimal UMP impact on absenteeism.

DISCUSSION

Major Findings

- Largest effects observed among female, male, Asian, White, and Black students in high-need areas
- Foster youth did not benefit, highlighting need for targeted support beyond food access

California’s UMP significantly reduces chronic absenteeism across demographics, with the strongest effects observed in high poverty districts.

Future Implications

- Encourage expanding UMP to other states
- Inform policies targeting absenteeism reduction and long-term crime prevention

Future Research

- Examine effects of UMP on academic and non-academic outcomes (e.g., mental health, nutrition, crime)
- Differentiate reasons for foster care entry (e.g., abuse, neglect, parental substance use) and consider time spent in care to better understand increased absenteeism
- Heterogeneity by grade level and geographic location

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